

FUNCTIONAL SPECIFICATION DOCUMENT (FSD)

Project Title

SCRIPT-CONTROLLED ACL – RESTRICT RECORD ACCESS BASED ON FIELD VALUE ASSIGNED

1. Document Purpose

This Functional Specification Document describes the functional requirements and access-control behaviour of the ServiceNow project “Script-Controlled ACL – Restrict Record Access Based on Field Value.”

The project demonstrates how ServiceNow Access Control Lists (ACLs) can be used to control record-level access based on user roles and the value of a record field.

In this project, users belonging to the EEE branch are allowed to view records where the Branch field is set to EEE, while administrators retain full access.

2. Project Objective

The main objective of the project is to implement record-level security using ServiceNow ACLs.

The system is designed to:

- Create users and custom roles.
- Create a custom Institution Details table.
- Store student-related information.
- Configure branch values such as ECE, EEE, and CSE.

- Restrict Read access using an ACL.
- Use a Data Condition based on the Branch field.
- Use a script to evaluate user roles.
- Provide Create, Write, and Delete permissions through separate ACLs.
- Prevent unauthorized users from accessing protected records.
- Provide administrators with full access.

The project demonstrates how roles and record field values can be evaluated together to make access decisions.

3. Scope of the System

The scope of this project covers the implementation and testing of ACL-based security for the custom Institution Details table.

The system includes:

1. User creation.
2. Custom role creation.
3. Role assignment.
4. Custom table creation.
5. Field configuration.
6. Student record creation.
7. Read ACL configuration.
8. Create ACL configuration.
9. Write ACL configuration.
10. Delete ACL configuration.
11. Script-controlled access.
12. Verification through user impersonation.

The project specifically focuses on record-level access control in ServiceNow.

4. System Overview

The system uses ServiceNow's Access Control List mechanism to determine whether a user can perform an operation on a record.

The access-control flow is:

User → Assigned Role → ACL Evaluation → Field/Data Condition → Script Evaluation → Allow/Deny Access

For Read access, the ACL is configured for the "u_institution_details" table with the bb1 role and a Data Condition where Branch is EEE.

The system also provides separate ACLs for Create, Write, and Delete operations using the roles bb2, bb3, and bb4 respectively.

5. User Roles

5.1 Administrator

The administrator has full access to the records.

The Read ACL script explicitly allows users with the "admin" role to access the records.

Administrator Functions

- View records.
- Access records regardless of branch.

- Perform administrative operations according to the configured ACLs.

5.2 EEE User – bb1

The test user is created as EEE User and is assigned the custom role bb1.

The bb1 role is used for Read access.

Function

- View permitted EEE branch records.

The verification procedure specifies that a user with the bb1 role can view only EEE branch records.

5.3 Create Role – bb2

The bb2 role is used for Create access.

A user having the required roles can access the New functionality for creating records.

5.4 Write Role – bb3

The bb3 role is used for Write access.

Users with the required combination of roles can view and edit permitted EEE branch records.

5.5 Delete Role – bb4

The bb4 role is used for Delete access.

Users with the required combination of roles can perform the Delete operation on permitted records.

6. Functional Requirements

FR-01: User Creation

The system shall allow an administrator to create the test user EEE User.

The user details specified in the project are:

- User ID: EEE User
- First Name: EEE
- Last Name: User
- Email: eeeuser@gmail.com

The user is created through User Administration → Users.

FR-02: Role Creation

The system shall allow creation of the following custom roles:

- bb1
- bb2
- bb3
- bb4

The roles shall be assigned to the EEE User as required.

FR-03: Institution Details Table

The system shall provide a custom table named:

Label: Institution Details

Table Name: "u_institution_details"

The table shall not extend another table according to the project configuration.

FR-04: Field Management

The Institution Details table shall contain the following fields:

Field| Type

Student Roll Number| Auto Number

Student Name| Reference – User

Faculty Name| Reference – User

Branch| Choice

Email| String

Phone Number| String
Description| Multi String

The Branch field shall contain:

- ECE
- EEE
- CSE

FR-05: Record Creation

The system shall contain multiple student records with different branch values.

The project specifies records for:

- ECE
- EEE
- CSE

These records are required to verify the ACL behaviour.

7. Access Control Requirements

7.1 Read ACL

The Read ACL shall be configured with:

Property| Value

Type| Record
Operation| Read
Name| "u_institution_details"
Active| True
Advanced| True
Required Role| bb1
Data Condition| Branch is EEE

The Security_admin role must be elevated before creating the ACL.

7.2 Read ACL Script

The Read ACL shall use a script to evaluate the user's role.

The specified script is:

```
(function () {  
  
    // Allow admin users full access  
    if (gs.hasRole('admin')) {  
        return true;  
    }  
  
    // Allow only EEE branch users to see EEE records  
    if (gs.hasRole('bb1')){  
        return true;  
    }  
  
    // Deny access for all others  
    return false;  
})
```

```
})();
```

The script allows administrators, allows users having the "bb1" role, and denies access for other users.

8. Create ACL

The Create ACL shall be configured as follows:

Property	Value
Type	Record
Operation	Create
Name	"u_institution_details"
Active	True
Required Role	bb2
Data Condition	Not required

The ACL controls the Create operation for the Institution Details table.

9. Write ACL

The Write ACL shall be configured as follows:

Property	Value
Type	Record
Operation	Write
Name	"u_institution_details"
Active	True
Required Role	bb3

Data Condition| Not required

The Write ACL allows the required users to edit permitted records.

10. Delete ACL

The Delete ACL shall be configured as follows:

Property| Value

Type| Record

Operation| Delete

Name| "u_institution_details"

Active| True

Required Role| bb4

Data Condition| Not required

The Delete ACL controls deletion of permitted records.

11. Functional Workflow

The complete system workflow is:

Step 1

Administrator logs in to the ServiceNow instance.

Step 2

The administrator creates the EEE User.

Step 3

Custom roles bb1, bb2, bb3, and bb4 are created and assigned.

Step 4

The Institution Details table is created.

Step 5

Required fields are added to the table.

Step 6

Student records are created with different branch values.

Step 7

The Read ACL is configured with the bb1 role and the EEE branch condition.

Step 8

The Read ACL script evaluates the user's role.

Step 9

Create, Write, and Delete ACLs are configured using bb2, bb3, and bb4.

Step 10

Different users are impersonated to verify the access behaviour.

The PDF specifies verification using the EEE user and Admin user through the Student Records list.

12. Business Rules

The following access rules are implemented:

Rule 1

An administrator shall have full access to records.

Rule 2

A user with the bb1 role shall be allowed Read access according to the configured ACL and EEE branch condition.

Rule 3

A user without the required role shall not be allowed to access protected records.

Rule 4

The bb2 role controls Create access.

Rule 5

The bb3 role controls Write access.

Rule 6

The bb4 role controls Delete access.

Rule 7

Create, Write, and Delete ACLs do not require a Data Condition according to the project specification.

13. Non-Functional Requirements

13.1 Security

The system shall prevent unauthorized users from viewing, modifying, creating, or deleting protected records.

13.2 Access Control

Access shall be controlled through ServiceNow roles and ACL configurations.

13.3 Data Protection

Sensitive records shall be protected from unauthorized operations.

13.4 Maintainability

Different operations are separated into different ACL configurations, making the access rules identifiable by operation.

The project outcome specifically describes protection against unauthorized viewing, modification, creation, and deletion across forms, lists, and Service Portal interfaces.

14. Verification and Testing

The system shall be tested by impersonating different users and opening the Student Records list.

Test Case 1 – Administrator Read Access

User: Admin

Operation: Read

Expected Result: Administrator can view all records regardless of branch.

Test Case 2 – bb1 Read Access

User: EEE User with bb1

Operation: Read

Expected Result: User can view the permitted EEE branch records.

Test Case 3 – User Without Required Role

User: User without required role

Operation: Read

Expected Result: Access is denied.

Test Case 4 – Create Access

Roles: bb1 + bb2

Operation: Create

Expected Result: New button/Create functionality is available.

Test Case 5 – Write Access

Roles: bb1 + bb2 + bb3

Operation: Write

Expected Result: User can view and edit the permitted EEE branch records.

Test Case 6 – Delete Access

Roles: bb1 + bb2 + bb3 + bb4

Operation: Delete

Expected Result: User can view, create, edit, and delete the permitted EEE branch records.

The PDF defines these progressive verification behaviours for the ACLs.

15. Error and Access-Denial Behaviour

If a user does not satisfy the required ACL conditions, the corresponding operation shall not be permitted.

For the Read ACL, the script returns "false" for users who do not satisfy the specified role condition.

This prevents unauthorized users from accessing protected records.

16. Security Considerations

The security of the system depends on:

- Correct role assignment.

- Correct ACL configuration.
- Correct Data Condition configuration.
- Correct script implementation.
- Proper elevation of the Security_admin role during ACL creation.
- Testing with different users.

The Read ACL requires the Security_admin role to be elevated during configuration.

17. Assumptions

The following assumptions are based on the project specification:

- The ServiceNow instance is available.
- The administrator has the required administrative access.
- The Security_admin role can be elevated for ACL configuration.
- The custom table "u_institution_details" is available after creation.
- Student records contain branch values such as ECE, EEE, and CSE.
- The required custom roles are created before ACL testing.

18. Expected Outcome

After successful implementation, the system shall demonstrate record-level security using ServiceNow ACLs.

The project shall demonstrate:

- Role-based Read access.
- Branch-based record restriction.
- Script-controlled access.

- Create access using bb2.
- Write access using bb3.
- Delete access using bb4.
- Full administrator access.

The final outcome is an understanding of how READ, WRITE, CREATE, and DELETE ACLs enforce record-level security and how user roles and record field values are evaluated together to make access decisions.

19. Conclusion

The Script-Controlled ACL – Restrict Record Access Based on Field Value project provides a functional implementation of record-level security in ServiceNow.

The system creates users and custom roles, stores student information in the Institution Details table, and uses ACLs to control Read, Create, Write, and Delete operations.

The Read ACL combines the required role, Branch data condition, and script-based evaluation. Separate ACLs are then used to control Create, Write, and Delete operations.

Thus, the functional design demonstrates how ServiceNow ACLs can protect records from unauthorized access and operations. Idhu FSD-ku proper content da. Nee already prepare pannina documentation-la irukkura same project details-a base pannithaan structure pannirukken; PDF-la irukkadha extra technical requirements-a invent pannala.

If your college has a specific FSD format/template (for example Purpose → Scope → Functional Requirements → Use Case → Inputs/Outputs → Acceptance Criteria), upload that template and I can fit this exact project into it.